

# Statistical Analysis of Electric Vehicle Charging Load Distribution Using Latin Square Design

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**Abstract**—The rapid growth of electric vehicles (EVs) has increased the load on electric vehicle charging stations. Charging energy data often shows high variability because they are influenced by external factors such as charging time and station location. This makes it difficult to clearly identify the effect of charging power levels. This study compares three Design and Analysis of Experiments approaches, which are Completely Randomized Design (CRD), Randomized Complete Block Design (RCBD), and Latin Square Design (LSD). The analysis uses EV charging transaction data from Boulder, Colorado. Charging power levels are treated as the main factor, while time-of-day and station location are used as blocking factors. Analysis of Variance (ANOVA) results show that charging power levels have a significant effect on total energy consumption. The Latin Square Design produces the lowest error variance compared to CRD and RCBD, indicating higher efficiency. Post-hoc analysis shows that low-power charging uses significantly less energy than higher charging levels. These results suggest that Latin Square Design is an effective method for analyzing EV charging data with high variability and can support better energy management at charging stations.

**Keywords**—Electric Vehicle Charging, Energy Consumption, Latin Square Design, Design of Experiments, ANOVA

## I. INTRODUCTION

### A. Background

Electric Vehicles (EVs) have experienced rapid global development in recent years, driven by the urgent need for sustainable transportation and carbon emission reduction [1]. As the number of electric cars increases, charging stations have evolved into a critical component of grid load management [2]. Analyzing electrical energy consumption within this infrastructure is crucial, not only for ensuring adequate capacity for vehicle charging but also for understanding the efficiency of energy transfer across various power levels (Grid Load Management) [3]. In addition, understanding how often people charge is crucial so that the distribution of electric current to charging stations can be regulated to optimize the community's charging frequency management strategy [4].

However, empirical data obtained from Open Data Boulder Colorado exhibits high variability. The response variable (Energy\_kWh) is influenced not only by the primary factor specifically the port type but also by extraneous factors such as station location and charging durations. Studies show that user behavior patterns, such as time, introduce stochastic noise that complicates this analysis [5], [6]. Without proper statistical control, the variability arising from these external

factors can lead to biased conclusions. For instance, charging patterns in city centers during the day may differ significantly from those in residential areas at night, resulting in inaccurate analytical outcomes [7].

In statistical science, Design and Analysis of Experiments (DAE) provides a robust framework for isolating treatment effects from environmental variability. Unlike purely observational studies, Design and Analysis of Experiments (DAE) allows researchers to control confounding factors to minimize experimental error variance [8]. This study involves comparing distinct experimental design approaches, including Completely Randomized Design (CRD), Randomized Complete Block Design (RCBD), and Latin Square Design (LSD) [9].

This research aims to compare these three methods (CRD, RCBD, and LSD) utilizing Electric Vehicle Charging Station Data from Open Data Boulder Colorado. The primary objective is to evaluate which method is most effective in controlling the variability caused by station location and charging times. Through Analysis of Variance (ANOVA), this study intends to accurately quantify the effect of charging power levels on total energy consumption and justify the selection of the most appropriate design for the analysis of electric vehicle charging [10].

### B. Research Objective

Based on the problems, this research aims to:

- Comparative analysis of three Design and Analysis of Experiments approaches: Completely Randomized Design (CRD), Randomized Complete Block Design (RCBD), and Latin Square Design (LSD), using datasets from Open Data Boulder, Colorado.
- Evaluate the effectiveness of each approach in controlling variability caused by specific factors and justify the selection of the most robust model.
- Apply the optimal approach using the Analysis of Variance (ANOVA) method to measure the effect of charging power levels on energy consumption efficiency.

### C. Benefits

This research is expected to provide significant results in contributing to the development of scientific knowledge and practical application in the field:

- **Evaluation of Approach Effectiveness:** Able to explain and provide a comparison of approaches between Completely Randomized Design (CRD), Randomized Complete Block Design (RCBD), and Latin Square Design (LSD), which can serve as a reference for other researchers in selecting a Design and Analysis of Experiments approach to solve problems related to specific factors.
- **Validation of Methodology on Observational Data:** Provides concrete evidence of the use of the Design and Analysis of Experiments approach in real-world situations with high variability.
- **Optimization of Energy Management at Charging Stations:** Provides insights for grid managers at charging stations regarding efficient and accurate charging based on power levels.
- **EV Infrastructure Planning:** Assists infrastructure planners in determining the most strategic locations for new charging stations based on the analysis of the impact of location on energy consumption.

## II. RESEARCH METHODOLOGY

### A. Data Source

The analysis relies on a secondary dataset comprised of transaction records from electric vehicle (EV) charging stations. This data covers sessions at both public and semi-public facilities. For every observation, the dataset provides specific session details, including the station identifier, the start time, the total duration of the session, and the aggregate energy dispensed (measured in kWh).

### B. Data Preprocessing and Feature Engineering

Preprocessing was undertaken to ensure the consistency and validity of the data for experimental analysis. Charging start times were standardized into datetime objects, while charging durations, originally recorded in hours, minutes, and seconds, were converted into a total hourly format. Charging power (kW) was derived by dividing the total energy consumed by the duration of the session.

Invalid observations, including missing timestamps, zero or negative charging durations, and non-positive energy values, were removed. To facilitate blocking, charging sessions were categorized into four time-of-day blocks based on their start hour: morning (06:00–11:59), afternoon (12:00–17:59), evening (18:00–23:59), and night (00:00–05:59).

Charging station locations were classified into four functional categories - commercial, workplace, public, and residential - based on station name patterns. Charging power was discretized into four treatment levels: low (<3.0 kW), standard (3.0–4.99 kW), high (5.0–6.99 kW), and ultra (≥7.0 kW).

### C. Comparative Efficiency Evaluation

The effectiveness of the Latin Square Design was evaluated by comparing its error mean square against estimated error variances from alternative designs, specifically the Completely Randomized Design (CRD) and

Randomized Complete Block Designs (RCBD) with single blocking factors. Relative efficiency was calculated as the ratio of the estimated mean square error of the alternative design to that of the Latin Square Design.

### D. Experimental Design

A Latin Square Design (LSD) was employed to evaluate the effect of charging power levels on energy consumption while controlling two nuisance factors: time-of-day and location type. The design consists of a 4×4 matrix, where rows represent time blocks, columns represent location categories, and treatments correspond to charging power levels.

This structure ensures orthogonality, as each power level appears exactly once within every row and column. Because the observational nature of the data did not naturally fit a perfectly balanced experiment, we aggregated the mean energy consumption for every combination of time, location, and power. In instances where a design cell lacked observations, we maintained the design's integrity by inputting the mean value of the corresponding treatment level.

### E. Statistical Model

The following additive linear model was used for analysis:

$$Y_{ijk} = \mu + R_i + C_j + T_k + \varepsilon_{ijk},$$

where  $Y_{ijk}$  denotes the average energy consumed,  $\mu$  is the overall mean,  $R_i$  represents the effect of the  $i$ -th time block,  $C_j$  represents the effect of the  $j$ -th location category,  $T_k$  represents the effect of the  $k$ -th charging power level, and  $\varepsilon_{ijk}$  is the random error term assumed to be independently and normally distributed with constant variance.

### F. Statistical Analysis

Analysis of variance (ANOVA) was conducted to assess the significance of charging power level effects while accounting for time and location blocks. Model assumptions were evaluated using the Shapiro Wilk test for normality of residuals and Levene's test for homogeneity of variances.

When a significant treatment effect was detected, post-hoc comparisons were performed using Tukey's Honestly Significant Difference (HSD) test for conservative inference and Fisher's Least Significant Difference (LSD) test for more sensitive pairwise comparisons.

## III. RESULTS AND DISCUSSION

### A. Descriptive Results of the Latin Square Structure

After data preprocessing and aggregation, a Latin Square Design was successfully constructed using four time-of-day blocks and four charging station location categories. Each cell in the design represents the mean energy consumption associated with a specific charging power level. This structure ensures that each treatment appears exactly once within every row and column, allowing the effects of time-of-day and location to be systematically controlled.

Table I. Latin Square Table

| Row Time  | Col Loc     | Treatment  | Energy kWh |
|-----------|-------------|------------|------------|
| morning   | Commercial  | A Low      | 4.157124   |
| morning   | Public      | B Standard | 8.766236   |
| morning   | Workplace   | C High     | 12.213720  |
| morning   | Residential | D Ultra    | 13.170936  |
| afternoon | Commercial  | B Standard | 5.845340   |
| afternoon | Public      | C High     | 10.913229  |
| afternoon | Workplace   | D Ultra    | 14.159724  |
| afternoon | Residential | A Low      | 3.686677   |
| evening   | Commercial  | C High     | 11.665284  |
| evening   | Public      | D Ultra    | 14.159724  |
| evening   | Workplace   | A Low      | 5.826298   |
| evening   | Residential | B Standard | 6.768876   |
| night     | Commercial  | D Ultra    | 14.159724  |
| night     | Public      | A Low      | 4.905358   |
| night     | Workplace   | B Standard | 22.890636  |
| night     | Residential | C High     | 14.906348  |

Table I summarizes the resulting Latin Square, where charging power levels (A Low, B Standard, C High, and D Ultra) are evenly distributed across both blocking factors. The balanced arrangement satisfies the orthogonality property of the Latin Square Design and minimizes confounding between treatment and block effects. As a result, the observed energy consumption values can be interpreted more clearly in relation to charging power levels.

The average energy consumption values in Table I reveal a consistent increasing trend as charging power levels rise. Low-power charging sessions (A Low) exhibit the lowest mean energy consumption across different times and locations, while ultra-fast charging (D Ultra) shows the highest average energy usage. The standard and high charging levels fall between these two extremes, indicating a gradual increase in energy consumption as charging power increases.

This descriptive pattern suggests that charging power levels play an important role in determining total energy usage. Although these results are descriptive in nature, they provide an initial indication of treatment effects. Therefore, further inferential analysis using Analysis of Variance (ANOVA) is required to formally assess the statistical significance of these observed differences.

#### B. Comparison of Experimental Design Approaches

To address the first research objective, the performance of different experimental designs was evaluated through a comparative efficiency analysis. Estimated mean square errors (MSE) were derived for the Completely Randomized Design (CRD), Randomized Complete Block Designs (RCBD), and the Latin Square Design (LSD). Relative efficiency values were calculated to quantify how well each design-controlled variability in the dataset.

Table II. Efficiency Comparison Table

| Desain          | Relative Efficiency |
|-----------------|---------------------|
| CRD             | 1.46                |
| CBD (Duration)  | 1.22                |
| RCBD (Location) | 1.39                |
| Latin Square    | 1.00                |

The CRD exhibited the highest estimated error variance, with a relative efficiency of 1.46. This result indicates that ignoring both time-of-day and location effects leads to substantial unexplained variability in energy consumption. It confirms that EV charging data is highly heterogeneous and

unsuitable for analysis under a fully randomized experimental structure.

Introducing a single blocking factor through RCBD improved statistical efficiency. RCBD with time-of-day as the block achieved a relative efficiency of 1.22, while RCBD with station location as the block yielded a value of 1.39. Although both approaches reduced error variance compared to CRD, they were unable to simultaneously control temporal and spatial variability.

The Latin Square Design demonstrated the highest efficiency, serving as the reference with a relative efficiency of 1.00. This confirms that LSD is the most appropriate design for this dataset, as it effectively accounts for both nuisance factors at the same time. Consequently, the treatment effect of charging power level can be estimated with greater precision.

The use of a Graeco-Latin Square Design (GLSD) was not considered, as the dataset does not contain a valid third nuisance factor. Introducing an artificial or dummy factor would violate experimental assumptions and reduce the available degrees of freedom. Furthermore, a Balanced Incomplete Block Design (BIBD) was deemed unnecessary because a complete 4×4 experimental layout was successfully constructed, making incomplete block methods redundant.

#### C. ANOVA Results for the Latin Square Design

Table III. ANOVA TABLE (Main Results)

|           | df | Sum Sq | Mean Sq | F value | Pr(>F) |
|-----------|----|--------|---------|---------|--------|
| Row Time  | 3  | 75.50  | 25.17   | 2.165   | 0.193  |
| Col Loc   | 3  | 58.02  | 19.34   | 1.664   | 0.272  |
| Treatment | 3  | 199.86 | 66.62   | 5.732   | 0.034  |
| Residuals | 6  | 69.73  | 11.62   |         |        |

The ANOVA results obtained from the Latin Square Design model are presented in Table III. The analysis indicates that the charging power level (Treatment) has a statistically significant effect on energy consumption, with an F-value of 5.732 and a p-value of 0.034. This result demonstrates that differences in charging power levels lead to meaningful variations in total energy usage.

In contrast, the effects of time-of-day (row factor) and location category (column factor) were not statistically significant. The p-values for these factors were 0.193 and 0.272, respectively, which are both above the 0.05 significance level. This indicates that, after applying the blocking structure, variability associated with time and location is adequately controlled and does not confound the treatment effect.

The significance of the treatment effect directly supports the research objective of identifying the most influential factor affecting energy consumption. By controlling two major nuisance variables simultaneously, the Latin Square Design allows the treatment effect to be estimated with greater precision. As a result, the observed differences in energy consumption can be attributed primarily to charging power levels rather than external temporal or spatial factors.

The diagnostic results confirm that the key assumptions of the ANOVA model are adequately met. This indicates that the normality and homogeneity of variance assumptions required for valid ANOVA inference are satisfied. Consequently, the inferential results obtained from the model can be interpreted with confidence.

#### D. Assumption Validation

The assumptions underlying the Analysis of Variance (ANOVA) model were evaluated to ensure the validity of the inferential results obtained from the Latin Square Design. In this study, the primary assumptions assessed were the normality of residuals and the homogeneity of variances. These assumptions are essential to ensure that the F-test results from the ANOVA are statistically reliable.

##### 1) Normality Test

The normality of the model residuals was assessed using the Shapiro-Wilk test. The test produced the following results:

$$W = 0.91768, \quad p = 0.1548$$

Since the p-value exceeds the 0.05 significance level, the null hypothesis that the residuals follow a normal distribution cannot be rejected. This indicates that the residuals are approximately normally distributed. Therefore, the normality assumption required for ANOVA is satisfied.

##### 2) Homogeneity of Variance Test

To evaluate the assumption of equal variances across treatment groups, Levene's test was applied. The test result is summarized as follows:

$$F = 1.361, \quad p = 0.3015$$

Because the p-value is greater than 0.05, the null hypothesis of equal variances cannot be rejected. This suggests that the variability of energy consumption is statistically homogeneous across charging power levels.

Overall, the diagnostic results confirm that the key assumptions of the ANOVA model are adequately met. Since both normality and homogeneity of variance assumptions hold, the conclusions drawn from the Latin Square ANOVA are statistically valid. Consequently, inferential results can be interpreted with confidence.

#### E. Post-Hoc Analysis of Charging Power Levels

Following the identification of a significant treatment effect in the ANOVA results, post-hoc analyses were conducted to determine specific differences among charging power levels. Two post-hoc methods were applied, namely Tukey's Honestly Significant Difference (HSD) test and Fisher's Least Significant Difference (LSD) test. The use of both methods allows for a balanced interpretation by comparing conservative and more sensitive inferential approaches.

Table IV. Post-Hoc Test 1: Tukey HSD

|                    | diff     | lwr       | upr       | p adj     |
|--------------------|----------|-----------|-----------|-----------|
| B_Standard-A_Low   | 6.423908 | -1.920832 | 14.768648 | 0.1287471 |
| C_High-A_Low       | 7.780781 | -0.563959 | 16.125521 | 0.0656731 |
| D_Ultra-A_Low      | 9.268663 | 0.923923  | 17.613403 | 0.0323914 |
| C_High-B_Standard  | 1.356873 | -6.987867 | 9.701613  | 0.9395688 |
| D_Ultra-B_Standard | 2.844755 | -5.499985 | 11.189495 | 0.6594215 |

|                |          |           |          |           |
|----------------|----------|-----------|----------|-----------|
| D_Ultra-C_High | 1.487882 | -6.856858 | 9.832622 | 0.9229010 |
|----------------|----------|-----------|----------|-----------|

Table IV presents the results of Tukey's HSD test. The analysis reveals a statistically significant difference between the ultra-fast charging level (D\_Ultra) and the low-power charging level (A\_Low), with an adjusted p-value of 0.032. Other pairwise comparisons did not reach statistical significance, which is consistent with the conservative nature of Tukey's method in controlling the family-wise error rate.

Table V. Post-Hoc Test 2: Fisher's LSD

|            | Energy kWh | groups |
|------------|------------|--------|
| D Ultra    | 13.912527  | a      |
| C High     | 12.424645  | a      |
| B Standard | 11.067772  | a      |
| A Low      | 4.643864   | b      |

In contrast, the results of Fisher's LSD test, shown in Table V, provide a more sensitive comparison among treatment levels. The test separates the low-power charging level (A\_Low) into a distinct group, while the standard, high, and ultra-fast charging levels are grouped together. This indicates that low-power charging consumes significantly less energy than higher charging power levels, whereas differences among the higher levels are not statistically significant.

These findings suggest that the most substantial increase in energy consumption occurs when transitioning from low-power to higher-power charging. Incremental increases among standard, high, and ultra-fast charging levels are comparatively smaller and less distinguishable. These results provide further insight into how charging power levels influence energy usage and support the practical interpretation of the ANOVA findings.

#### F. Visualizations and Interpretation

Figure 1. Energy Consumption by Charging Speed

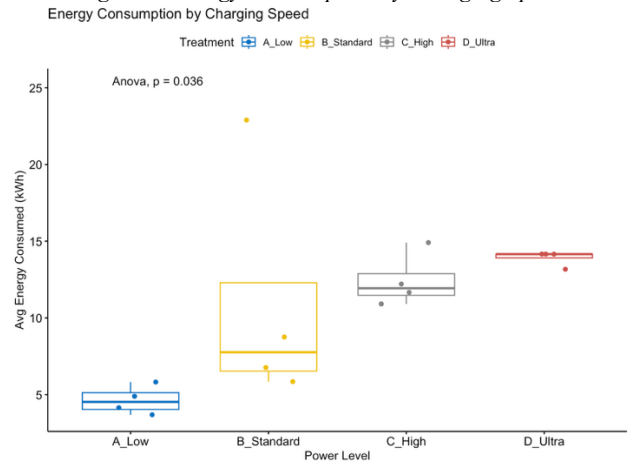


Figure 1 illustrates the distribution of average energy consumption across different charging power levels using a boxplot representation. The visualization provides a clear overview of the central tendency and variability of energy usage for each charging category. This graphical summary complements the numerical results obtained from the ANOVA and post-hoc analyses.

The boxplot shows a clear separation between the low-power charging level (A\_Low) and the higher charging power levels. The A\_Low group exhibits a substantially lower median energy consumption compared to the standard, high, and ultra-fast charging categories. In contrast, the D\_Ultra category displays the highest median value with relatively low dispersion, indicating more consistent energy usage at higher charging power levels.

The overall ANOVA p-value displayed in the figure ( $p = 0.036$ ) further supports the statistical significance of charging power level effects on energy consumption. This visual confirmation aligns with the inferential results, reinforcing the conclusion that charging power level is a key determinant of total energy usage. Therefore, the visualization strengthens the interpretation of the statistical findings by providing an intuitive representation of the observed treatment differences.

### G. Discussion

The results of this study demonstrate that charging power level is the primary factor influencing total energy consumption in electric vehicle charging sessions. Although raw charging data exhibits substantial variability due to temporal and spatial factors, these sources of variability can be effectively controlled through an appropriate experimental design. The findings confirm that once time-of-day and location effects are accounted for, the impact of charging power level becomes more clearly identifiable.

The comparison of experimental design approaches highlights the importance of selecting a design that aligns with the structure of the data. The Completely Randomized Design proved insufficient for handling the heterogeneity inherent in EV charging data, while Randomized Complete Block Designs offered partial improvements by controlling a single nuisance factor. In contrast, the Latin Square Design achieved superior efficiency by simultaneously controlling both time-of-day and location, resulting in reduced error variance and improved precision in estimating treatment effects.

From a practical perspective, the post-hoc analysis provides meaningful insights for energy management at charging stations. The significant difference between low-power charging and higher charging levels suggests that encouraging low to standard charging speeds may help reduce total energy consumption, particularly in contexts where charging duration is not a critical constraint. Meanwhile, the relatively small differences among higher charging power levels indicate diminishing returns in energy efficiency as charging speed increases.

This study emphasizes the critical role of experimental design in analyzing complex observational datasets. By applying the Latin Square Design, the analysis successfully isolates the effect of charging power levels while mitigating the influence of external variability. These findings support the use of structured experimental approaches for improving the reliability of statistical inference in electric vehicle charging studies.

## IV. CONCLUSION

This study demonstrates that charging power level has a significant effect on total energy consumption in electric

vehicle charging sessions. Although charging data is influenced by time-of-day and station location, these sources of variability can be effectively controlled using a Latin Square Design. Compared to the Completely Randomized Design and Randomized Complete Block Designs, the Latin Square Design provides higher statistical efficiency by simultaneously accounting for two nuisance factors.

The ANOVA results confirm that charging power levels significantly influence energy consumption. Post-hoc analyses indicate that low-power charging consumes substantially less energy than standard, high, and ultra-fast charging levels, while differences among the higher charging categories are relatively small. These findings are supported by diagnostic tests, which show that the assumptions of normality and homogeneity of variance are satisfied.

The results highlight the importance of selecting an appropriate experimental design when analyzing complex and heterogeneous observational datasets. The use of the Latin Square Design enables more reliable estimation of treatment effects and clearer interpretation of charging behavior patterns. This approach is well suited for supporting data-driven decision-making in the analysis and management of electric vehicle charging systems.

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